

MRC Lecture 3 — Derivation Guide 1

The InfoNCE Loss: Derivation from the Boltzmann Selection Probability

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1. Motivation

The **InfoNCE loss** is the training objective for all Specificity Foundation Models. It was introduced by Oord et al. (2018) from an information-theoretic perspective: it maximizes a lower bound on the mutual information between representations. CLIP (Radford et al., 2021) adopted it for cross-modal contrastive learning between images and text. CALM (Lee et al., 2026) adopted it for antibody-antigen binding prediction.

In this guide, we provide a completely independent derivation from physics: InfoNCE is the **maximum likelihood estimator** of the Boltzmann selection probability — the convergence equation. The training objective is not selected from a menu of contrastive losses. It is derived from the governing equation of molecular binding.

2. Setup: The Selection Scenario

Physical scenario. An agent molecule (e.g., an antibody) encounters a pool of candidate targets. The pool contains one correct binding partner f (the cognate target) and M non-cognate competitors $\{c_1, c_2, \dots, c_M\}$. The agent must “select” a target to bind.

Embeddings. The agent has been encoded into a query vector q by the agent encoder. Each target j has been encoded into a key vector k_j by the target encoder. The compatibility score between the agent and target j is:

$$s_j = \frac{\text{sim}(q, k_j)}{\tau}$$

where $\text{sim}(q, k_j) = \hat{q} \cdot \hat{k}_j$ is the cosine similarity (dot product of L2-normalized embeddings, from Lecture 1) and $\tau > 0$ is the temperature parameter.

3. The Selection Probability

From the convergence equation. At thermal equilibrium, the probability that the agent selects target j from the pool follows the Boltzmann distribution (Lecture 1, Derivation Guide 5; proven unique in Lecture 6):

$$P(j/q) = \frac{\exp(s_j)}{\sum_{l \in \text{pool}} \exp(s_l)}$$

This is the softmax function applied to the compatibility scores — the convergence equation.

Applied to the training scenario. The pool consists of the correct target f and M competitors. The probability of the agent correctly selecting its cognate target is:

$$P(f/q) = \frac{\exp(s_f)}{\exp(s_f) + \sum_{i=1}^M \exp(s_{c_i})}$$

where $s_f = \text{sim}(q, k_f)/\tau$ is the score for the correct target and $s_{c_i} = \text{sim}(q, k_{c_i})/\tau$ is the score for competitor i .

Biological interpretation. In the immune system, this corresponds to clonal selection: a lymphocyte (agent) must discriminate its cognate foreign antigen (correct target f) from self-antigens (competitors $c_1, \dots, c_M$). The probability of correct selection is governed by the Boltzmann distribution over binding energies. The training objective should maximize this probability.

4. Maximum Likelihood Estimation

The likelihood. Given a training set of N agent-target binding pairs $\{(q_1, f_1), (q_2, f_2), \dots, (q_N, f_N)\}$, each with its own competitor pool, the joint likelihood of all correct selections is:

$$L = \prod_{i=1}^N P(f_i / q_i)$$

The log-likelihood. Taking the logarithm (which is monotonic, so maximizing the log-likelihood is equivalent to maximizing the likelihood):

$$\log L = \sum_{i=1}^N \log P(f_i / q_i)$$

The negative log-likelihood. In machine learning convention, we minimize a loss function. The loss is the negative average log-likelihood:

$$L = -\frac{1}{N} \sum_{i=1}^N \log P(f_i / q_i)$$

5. Substituting the Convergence Equation

Step 1. Substitute the selection probability from §3:

$$L = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(s_{f_i})}{\exp(s_{f_i}) + \sum_{j=1}^M \exp(s_{c_j})}$$

Step 2. In the contrastive training setup, the competitor pool for agent i is the set of all other targets in the mini-batch. For a batch of N pairs, agent i 's correct target is target i , and the competitors are targets $\{1, \dots, N\} \setminus \{i\}$. The pool is the entire batch.

Define the similarity matrix: $S_{ij} = \text{sim}(q_i, k_j) / \tau$ for all $i, j = 1, \dots, N$.

The correct pair for agent i is target i , so $s_{f_i} = S_{ii}$ (the diagonal entry).

Step 3. Substitute:

$$L_{\text{InfoNCE}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(S_{ii})}{\sum_{j=1}^N \exp(S_{ij})}$$

This is the **InfoNCE loss**. ■

6. Interpretation

What the loss computes. For each agent i in the batch:

1. Compute the similarity S_{ij} with every target j in the batch
2. Apply softmax to get the probability distribution over targets:

$$P(j / q_i) = \exp(S_{ij}) / \sum_k \exp(S_{ik})$$
3. Extract the probability of the correct target: $P(i / q_i) = \exp(S_{ii}) / \sum_k \exp(S_{ik})$
4. Take the negative log of this probability: $-\log P(i / q_i)$

Average over all agents in the batch.

What the loss minimizes. The loss is minimized when $P(i/q_i)=1$ for every agent — when every agent selects its correct target with probability 1. This requires $S_{ii} \gg S_{ij}$ for all $j \neq i$: the correct pair has much higher similarity than any incorrect pair.

Theoretical bounds:

- **Minimum:** $L_{\text{InfoNCE}}=0$ when $P(i/q_i)=1$ for all i (perfect retrieval).
 - **Maximum:** $L_{\text{InfoNCE}}=\log N$ when $P(i/q_i)=1/N$ for all i (random guessing — all targets are equally likely). This is because $-\log(1/N)=\log N$.
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7. InfoNCE as Cross-Entropy

Claim. The InfoNCE loss for agent i is equivalent to the cross-entropy between a one-hot label and the softmax distribution over targets.

Proof. The one-hot label for agent i is:

$$y_{ij} = \begin{cases} 1 & \text{if } j=i \\ 0 & \text{if } j \neq i \end{cases}$$

The predicted probability distribution is:

$$\hat{p}_{ij} = \frac{\exp(S_{ij})}{\sum_k \exp(S_{ik})} = \text{softmax}(S_{i,:})_j$$

The cross-entropy loss is:

$$\begin{aligned} H(y_i, \hat{p}_i) &= - \sum_{j=1}^N y_{ij} \log \hat{p}_{ij} = -1 \cdot \log \hat{p}_{ii} - \sum_{j \neq i} 0 \cdot \log \hat{p}_{ij} = -\log \hat{p}_{ii} \\ &= -\log \frac{\exp(S_{ii})}{\sum_k \exp(S_{ik})} \end{aligned}$$

This is exactly the InfoNCE loss for agent i . ■

Implication. InfoNCE is multi-class cross-entropy applied to the $N \times N$ similarity matrix, where the “classes” are the N targets and the correct class for agent i is target i (the diagonal). This is why InfoNCE training can be implemented efficiently using standard cross-entropy loss functions available in any deep learning framework, with the similarity matrix as logits and the diagonal indices as labels.

8. The Role of In-Batch Negatives

Definition. In the InfoNCE setup, the competitor pool for each agent is the entire mini-batch. For agent i , all targets $j \neq i$ serve as **in-batch negatives** — surrogate non-binding partners.

Why this works. In a randomly sampled mini-batch of N binding pairs, the probability that two randomly paired molecules (agent i with target $j \neq i$) actually bind is extremely small — there are millions of possible antigens for each antibody, so a random antigen is almost certainly a non-binder. In-batch negatives are therefore valid (though noisy) samples of the non-binding distribution.

Batch size as competitor pool size. The number of competitors $M = N - 1$ equals the batch size minus one. Larger batches provide more negatives, making the discrimination task harder and the learning signal richer. The relationship between batch size and learning quality is analogous to the relationship between competitor pool size and selection stringency in the Boltzmann distribution: more competitors require sharper discrimination.

CLIP used batch sizes up to 32,768. **CALM** used smaller batches (limited by the ~4,000 available pairs) but supplemented with hard negative mining strategies.

9. Connection to Mutual Information

The information-theoretic perspective. Oord et al. (2018) proved that InfoNCE maximizes a lower bound on the mutual information $I(q; k)$ between agent and target representations:

$$I(q; k) \geq \log N - L_{\text{InfoNCE}}$$

As the loss decreases (better discrimination), the lower bound tightens. At perfect performance ($L=0$), the bound equals $\log N$.

The physical perspective. The MRC framework provides a complementary interpretation: InfoNCE maximizes the likelihood of the Boltzmann selection probability. The two perspectives converge because the Boltzmann distribution IS the maximum entropy distribution (Jaynes, 1957), and maximum entropy distributions are precisely those that maximize mutual information subject to energy constraints. The information-theoretic and physical derivations are not merely analogous — they are different expressions of the same mathematical principle.

10. Summary

Concept	Definition	Significance
Selection probability	$P(f/q) = \exp(s_f) / \sum \exp(s_i)$	Boltzmann/softmax over competitor pool
InfoNCE loss	$-\frac{1}{N} \sum_i \log \frac{\exp(S_{ii})}{\sum_j \exp(S_{ij})}$	Negative log-likelihood of correct selection
In-batch negatives	All other targets in the mini-batch	Surrogate competitor pool
Cross-entropy equivalence	InfoNCE = cross-entropy with one-hot labels on S	Efficient implementation
Theoretical minimum	$L=0$	Perfect retrieval
Theoretical maximum	$L=\log N$	Random guessing
Mutual information	$I(q; k) \geq \log N - L$	Information-theoretic interpretation

11. Checkpoint Questions

Q1. In a batch of $N=256$ pairs, what is the InfoNCE loss if the model is guessing randomly (uniform distribution over targets)?

Answer: Random guessing means $P(i/q_i) = 1/N = 1/256$ for all i . The loss is

$-\log(1/256) = \log(256) = 8 \ln 2 \approx 5.55$ (in natural log) or $\log_2(256) = 8$ (in log base 2). This is the theoretical maximum.

Q2. The derivation assumes the convergence equation (softmax) as the selection probability. What happens if you start from a different selection function — say, the sigmoid — and derive the MLE?

Answer: Starting from the sigmoid $P(f) = \sigma(s_f) = 1/(1 + \exp(-s_f))$ gives binary cross-entropy:

$L = -\log \sigma(s_f) - \sum_{j \neq f} \log(1 - \sigma(s_j))$. This treats each pair independently — it does not model the full

competitor pool. In Lecture 6, we prove that sigmoid violates the IIA axiom, meaning it cannot correctly represent competitive molecular selection. The MLE of an incorrect model gives an incorrect loss function.

Q3. Why is the denominator of InfoNCE a sum over ALL targets in the batch, not just the negatives?

Answer: The denominator includes the correct target because it represents the full partition function of the Boltzmann distribution: $Z = \sum_{j \in \text{pool}} \exp(s_j)$. The pool includes the correct target — it is one of the

candidates the agent could bind. Excluding the correct target from the denominator would give

$P(f) = \exp(s_f) / \sum_{j \neq f} \exp(s_j)$, which does not sum to 1 over all candidates and is not a valid probability distribution.

12. Connection to Future Lectures

- **Lecture 3, WB2:** The symmetric loss averages InfoNCE computed in both directions (agent→target and target→agent), derived from the physical symmetry $\Delta G_{AB} = \Delta G_{BA}$.
- **Lecture 6:** The convergence equation — which we assumed as the starting point of this derivation — is proven to be the unique selection function satisfying five axioms. This closes the logical loop: the loss function is derived from the governing equation, and the governing equation is proven unique.
- **Lecture 7:** The full SFM architecture prescribes InfoNCE as the training objective. The derivation here shows it is the MLE of the convergence equation; Lecture 7 shows it is also the only loss consistent with all five architecture conditions.
- **Lecture 9:** The MRC scaling law predicts exponential convergence of retrieval accuracy with training data diversity. The exponential form follows from the fact that InfoNCE trains the model to estimate parameters in a known equation (the convergence equation), not to discover an unknown function.